

Lessons Learned from Video Game Players Sorting Genomes

Rogerio de Leon Pereira
Olivier Tremblay-Savard
deleonpr@myumanitoba.ca
olivier.tremblay-savard@umanitoba.ca
University of Manitoba
Winnipeg, Manitoba, CA

ABSTRACT

The fields of human computing and citizen science have made a lot of progress in creating engaging applications and encouraging people to donate part of their time to solve problems that are difficult for computers but easy for them. One option that has been widely used in citizen science particularly is to transform the problem into smaller tasks that can be presented as a puzzle game. This way, volunteers can participate by playing a game and contribute to scientific advancement. This work presents the lessons learned from the citizen science game GeSort, with a focus on better understanding player strategies and puzzle difficulty. Matches played over one year were analyzed. We examined which player strategies led to optimal and sub-optimal solutions. We analyzed how certain repetitive patterns of shapes and colours could confuse players and cause mistakes. Moreover, we present a simple system for dynamically adjusting the difficulty in a puzzle game when an achievable score is known. Based on our results, we propose some design guidelines that can generally be applied to any puzzle game in order to alleviate the potential issues related to puzzle presentation and patterns.

CCS CONCEPTS

• **General and reference** → *Design*; • **Applied computing** → **Computer games**.

KEYWORDS

GeSort, human computing, citizen science games, genome sorting problem, player strategies, puzzle difficulty, dynamic difficulty adjustment

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1 INTRODUCTION

Human computation emerged as a way to engage human participants in the process of solving problems that are naturally easier for them but still difficult for computers [26, 47]. An example of the applicability of human computing is reCAPTCHA, a system used to differentiate humans and computers interacting with web pages. This application had a dual function because it simultaneously identified users and helped with the digitization of books [50]. Another way to solicit the help of human participants is through games with a purpose [46]. At the same time that players have fun playing, they help solve a real world problem, as in the case of the ESP Game [48] in labelling images, or Peekaboom [49], in identifying objects in images.

Human computation can also be used to increase citizen participation in scientific development [17]. This happens by identifying bottlenecks where human capabilities can be focused, whether in curating images, classifying and organizing data, or solving puzzles. In some cases, this approach seems natural, as in Foldit, a citizen science game based on the problem of protein structure prediction, where players fold proteins [11]. In others, it is more subtle, as with the multiple sequence alignment problem presented in Phylo where the player participates by aligning colored blocks [22].

More recently, some researchers have used commercial games to increase the reach of their projects. The EVE online game, for example, is a commercial game, not a citizen science game. However, they added the *Project Discovery* to it, a series of tools that are not part of the core gameplay or game mechanics, but guarantees rewards within the main game environment for participating. Through these interactions, players help identify exoplanets and help with protein cataloging [28, 36]. A revamped version of Phylo was also recently inserted into the commercial game Borderlands 3, which led to a considerable increase in volunteer contributions to that project [52].

With all the recent applications of human computation in different domains, learning from past work contributes to the constant evolution of the field. In addition to the findings' importance and significance, sharing lessons learned during the process is imperative to guide the human computation and citizen science community [41].

This paper presents the lessons learned with GeSort [42], a citizen science game created to solve the genome sorting problem in bioinformatics, which is to find the shortest sequence of events that can transform one ordering of genes into another. As described below, the GeSort game presents the different genes using a combination of eight shapes and four colors, and the puzzles are centered

around a black dot, which represents an important biological location within the genomes. Moreover, GeSort was designed with a dynamic difficulty adjustement system, which separates all the puzzles into distinct difficulty levels and tracks the performance of the players to keep them in the most appropriate level.

In this paper, we conduct an analysis of the data collected over one year, between August 2021 and August 2022, to answer the following research questions:

(1) How are human players able to find more optimal solutions than those found by an algorithm?
(2) How does the location of the events in the puzzles affect player performance?
(3) Are there patterns that make the puzzles more difficult to solve?
(4) How is the dynamic difficulty adjustment system beneficial to the players and the GeSort game?

In our analysis of these questions, we discuss the different advantages and drawbacks of the GeSort game and provide general design guidelines that can be applicable to other similar human computation and citizen science applications.

2 LITERATURE REVIEW

2.1 Citizen Science and Games

Citizen science is scientific research carried out with the help and participation of the general public. The participants are usually not required to have any prior training in the domain area of the research project. The tasks that are entrusted to citizen scientists can involve collecting, classifying or curating data. The oldest documented citizen science project is the National Audubon Society's Christmas Bird Count, which started in 1900 [7]. Nowadays, there are several platforms on the Internet, such as Zooniverse[1] and SciStarter[2], where it is possible to connect volunteers with citizen science projects.

To expand the recruitment of volunteers, some citizen science projects have opted to gamify the tasks and convert them into full-fledged video games. This strategy allows to attract a broader base of participants, such as people who are more interested in the game itself and not necessarily the scientific project associated with it [52]. Another advantage of this approach is to increase engagement by turning a potentially repetitive activity into something fun, relaxing and rewarding. Citizens science games have been developed in many different areas, such as molecular biology [11, 22, 27], health sciences [1], astronomy [6, 8], deep learning [44] and neuroscience [23].

Developing citizen science games comes with its own challenges, such as how to guide the players towards good solutions when they are unknown [12], how game mechanics are affecting both player engagement and solution accuracy [31, 43], and how to address the barriers to expertise [32]. That is why it is essential for citizen science game developers to share the lessons learned during the design, development, testing and data collection stages of their games. Lessons related to the evaluation of puzzle hardness and solution quality [41], the use of touch screen and mouse interfaces [33], alternative scoring systems to increase engagement [15], or even the inclusion of educational content as a factor of engagement, retention and performance [14] can be used as guidelines for the creation of future successful citizen science games.

[1]https://www.zooniverse.org/
[2]https://scistarter.org/

2.2 Puzzle difficulty and visual patterns

In some cases, game elements can increase the difficulty unexpectedly. Visual elements can make the game more challenging in puzzle games or in games where the player needs to find and recognize patterns.

For example, humans more often scan the screen from left to right, top to bottom, in what is known as an F-Shaped Pattern [35]. Thus, players can choose the first option that appears, from left to right, without necessarily scanning the entire puzzle, which might make them miss a better move. Also, people tend to focus on objects or details that are more familiar, or objects that escape context, which is called visual salience [21]. Because of this, some patterns or pieces of the puzzle can accidentally draw the player's attention to the wrong region of the puzzle and make them miss the best possible moves.

Regarding how colors and shapes can confuse players, a possible explanation for this is *pareidolia*, the tendency to have a meaningful interpretation of a nebulous stimulus. As a result, it is possible to see a pattern, object, or meaning where there is none. A study suggests that pareidolia might be an important experience because the brain's process of identifying meaning in unknown or nonsense figures is very similar to identifying meaning in familiar objects [51].

Another possible explanation is the limited capacity of the human brain to process all the stimuli constantly captured. In human vision, for example, less than 1% of the visual input data is processed [54], which could lead to inattention blindness [16], when a person fails to perceive an unforeseen stimulus in plain sight. It can also be an optical or visual illusion: an illusion caused by the visual system and characterized by a visual perception that differs from reality [3, 4].

2.3 Dynamic Difficulty Adjustment (DDA)

Challenges and difficulty in games can vary in the form and the skill needed to overcome them, be it reaction time, reflex, coordination, mental or physical skills and problem-solving. However, such challenges need to be compatible with the players' skills. If the challenge is too high, the player feels frustrated; if it is too low, the player feels bored. The mental state where there is an ideal balance between the proposed challenge and the ability to overcome it is known as the flow state [5]. In game design, it is essential to adjust the game's difficulty to keep the player in that area between boredom and frustration, known as the flow channel [24, 39]. Balancing difficulty becomes challenging as each player has a different set of skills and reacts uniquely to the game's challenges. One strategy to resolve this dilemma and keep players in the flow channel is to adjust the game's difficulty dynamically [40, 55].

Strategies for dynamically adjusting difficulty in games are wide-ranging. One option is to change the demand for physical exertion, as presented in the *Nuts Catcher* game. In it, the player makes movements similar to those performed in physiotherapy sessions.

The game has a robot that changes the rigidity of the control to help or hinder the player's movement. The authors proposed a Q-Learning algorithm to alter game parameters to keep players in the flow. Their results indicated that the approach successfully modelled user behaviour to capture the adaptations and trends for each player as stated by the game's difficulties [2].

Another option is to change the sensory stimuli. One study used Gaussian Process Regression to change the in-game audio-cue setting in a first-person shooter game. The game changes the volume of enemy movement noise, making them easier or harder to detect. The results obtained by the authors suggest that the proposed approach can reach a recommended difficulty within a few plays [30].

A common strategy for implementing DDA is to adjust the game's difficulty once undesirable states such as boredom or frustration are observed. However, if there are no quantitative objectives, it is difficult to determine the appropriate level of intervention required to achieve the best configuration. One study proposed a DDA framework with the goal of global optimization to maximize player engagement throughout the game [53]. The authors modelled player progression as a probabilistic graph containing player states (nodes) and transition probabilities (edges). To maximize the engagement of players, the dynamic difficulty adjustment problem has been reduced to optimizing transition probabilities. The study applied this technique to several Electronic Arts games, showing an improvement of up to 9% in player engagement.

Another study used a parametric statistical model to describe the distribution of actions performed within a game level, thus creating a richer difficulty descriptor as an alternative to modelling the probability of success in completing levels. They tested the model on a dataset collected from the game Lily's Garden by Tactile Games. The results showed that the model can describe and explain the difficulty in most levels in a puzzle game, thus facilitating the choice of the level with the most suitable difficulty for each player [25].

In the context of a human computation game, the Glicko-2 player rating system (similar to the Elo rating system used in chess) was used to compare players with levels [37]. This was used initially to rate the difficulty of the different game levels with a first group of participants, and then in a matchmaking system that would constantly evaluate the strength of a player and assign the next level based on it. Although the authors did not observe an increase in participant engagement with the matchmaking system when compared to constantly increasing the difficulty level, they did observe that the matchmaking was more efficient in getting skilled players to work on harder tasks faster.

3 DESIGN OF THE CITIZEN SCIENCE GAME

In bioinformatics, comparing genomes is done to better understand evolutionary mechanisms. One type of comparison aims to infer the shortest sequence of evolutionary events capable of transforming one genome into another. This problem is known as the *genome sorting problem*. The genome sorting problem is usually solved on a *gene order*, in which every gene is represented by a label, and with a certain evolutionary model, describing which biological events are possible (e.g. duplications, deletions and inversions). The genome sorting problem is NP-hard in the presence of duplicate gene copies (*i.e.* when labels are reused) [9], and because of this, the existing algorithms for this problem are either heuristics [45] or they rely on solving integer linear programming equations which can be fairly slow [20]. Since the human brain is very good at recognizing patterns, GeSort was designed to help solve the genome sorting problem and analyze the players' strategies in an effort to design improved algorithms in the future [42].

3.1 GeSort: A Game for Sorting Genomes

GeSort (Figure 1) is a citizen science game created to be a broad and easy-to-access platform where people can share part of their time to solve genome sorting puzzles that are fun and relaxing. The game is currently available to be played on web browsers[3].

In the game, each puzzle is made up of two genomes and a top score, indicating (initially) the number of operations (moves) of a solution achieved by a heuristic. The top score gets updated each time a player achieves a better (lower) score. Through the available operations, the player's objective is to transform the bottom genome (mutable row) into the other (target row), preferably using fewer operations than the top score. There are three operations available in the game: a) duplication, where the player can copy one or more genes, b) deletion, where the player can delete one or more genes; and c) inversion, where the player can invert a sequence of one or multiple genes using the *terminus of replication* (the black dot in Fig. 1) as a pivot. The inversion event is allowed if and only if the selected genes contain the terminus or are right next to it, which represents how inversions usually occur in the studied genomes. Every time one or more genes are transported from one side of the terminus to the other, they are inverted. Thus, an arbitrary pair of genes A-B duplicated from one side of the genome will appear on the other as B-A.

The puzzles in the game are created based on bacterial genomes (more on that in section 3.2), which generally have only one circular DNA molecule (called a chromosome). In this type of molecule, replication begins in a region called the *origin of replication* and the progress of replication travels clockwise and counter-clockwise simultaneously, ending in an area known as *terminus of replication* [19, 34]. The specific types of genes that are being considered in GeSort are the transfer RNA (tRNA) and ribosomal RNA (rRNA) genes. In the genomes considered, there are 21 different types of tRNA genes (based on the amino acid they carry) and three types of rRNA genes, and they each are often present in multiple copies. Based on initial prototyping, we decided to use a combination of eight shapes that could be easily distinguished (empty square, filled square, circle, pentagon, hexagon, diamond, triangle and star) and four colors that are colorblind friendly (pink, orange, white and lime green) to represent the 24 gene types in the game. Each gene type was arbitrarily associated to a combination of a shape and a color. Note that some combinations (out of the 32 possibilities) are left unassigned so that we could add more types of genes in the future. In the game, the terminus of replication in the genomes is represented by a black circle, and both genomes are always aligned on the terminus. Each puzzle initially appears with the terminus centered on the screen. The player can move the view towards the left or right, but the genomes are always anchored on the terminus.

[3] https://gesort.cs.umanitoba.ca/

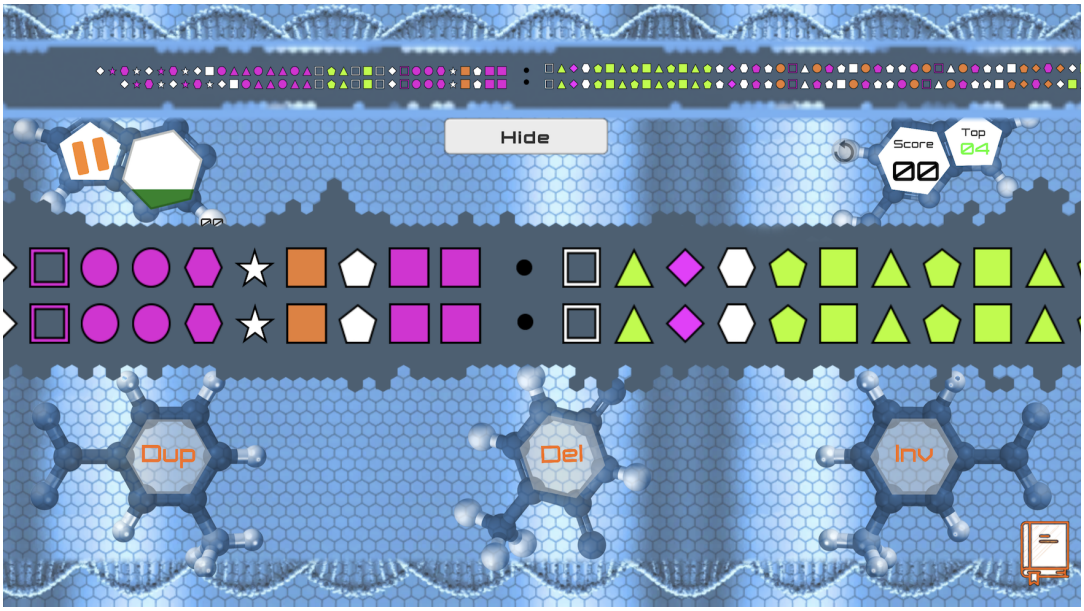


Figure 1: GeSort screenshot. In the middle are the two genomes (two rows of colored shapes). At the top of the image is the mini-map. The current score and the top score are displayed on the right. At the bottom are three buttons which correspond to the possible operations. The small white book in the lower right corner gives access to educational content.

3.2 The Puzzles

There are three types of puzzles, each created differently and with a specific role in the game's design. The first one is the group of tutorial puzzles. These were created manually by the research team and aimed to teach the player each possible game operation. The tutorial has six levels and is mandatory for every new player.

The second set of puzzles is the training puzzle set. They aim to train players and make them more familiar with the game. During the period of time analyzed in this work, there were 11 training levels. These puzzles were generated by a randomized generator script. The script receives the genome size and the number of evolutionary events as parameters. The script randomly creates a genome based on a gene bank (possible labels) and applies random evolutionary events to generate a new genome. These two genomes and the number of events make up a training puzzle, where the top score is the number of events used during the generation. These puzzles were relatively small, with genomes ranging from 6 to 15 genes and top scores ranging from 1 to 5. The intention was for players to move quickly through this stage.

After completing all the training puzzles, the player is introduced to the full game features, including the mini-map (an alternate representation that provides a global view of the genomes), the undo button, and the progression system. From that moment, the player only plays puzzles created with real genomes based on the phylogeny of *Clostridium* genomes, obtained from the PATRIC database [13]. An example of the *Clostridiales* order phylogeny is shown in Figure 2. According to preliminary studies, it was found that the *Clostridium* genomes are evolving more quickly than other bacterial genera, such as the *Bacillus* genus [10, 18, 29, 38], which makes these genomes interesting to study. Each puzzle was created with pairs of genomes that are closely related in the phylogeny to ensure a reasonable expected number of events. Mirrored versions of the selected puzzles (where the target and mutable genomes are swapped) were also created to test whether puzzles were easier in one direction or another, but no significant effect was observed. We used the OrthoAlign [45] algorithm to infer an evolutionary scenario between both genomes and used the number of events inferred to set the top score. The game was released with 64 puzzles created based on the *Clostridium* genus with genomes of 100 genes on average and the top score ranging from 1 to 36.

3.3 The Game Progression and the Dynamic Difficulty Adjustment

The first two stages of the game, tutorial and training, are linear. After these stages, players play puzzles based on real genomes, where the 64 puzzles were first ranked based on their top score and then divided into eight levels of difficulty, each containing eight puzzles. In other words, the eight puzzles with the lowest top score were assigned to difficulty level 1, the next eight to difficulty level 2 and so on. The number of difficulty levels (8) was selected to obtain a good separation between the top scores of different levels while maintaining a gradual progression in terms of difficulty. All players start the real genomes stage on difficulty level 1. However, the puzzles are randomly assigned in each difficulty level. That way, two players playing the same difficulty level would not play the puzzles of that difficulty in the same order. This design decision was made to ensure an even distribution of the number of puzzles played at each difficulty level.

When solving a puzzle, players receive progression points (which are hidden to the players) according to their performance. If the

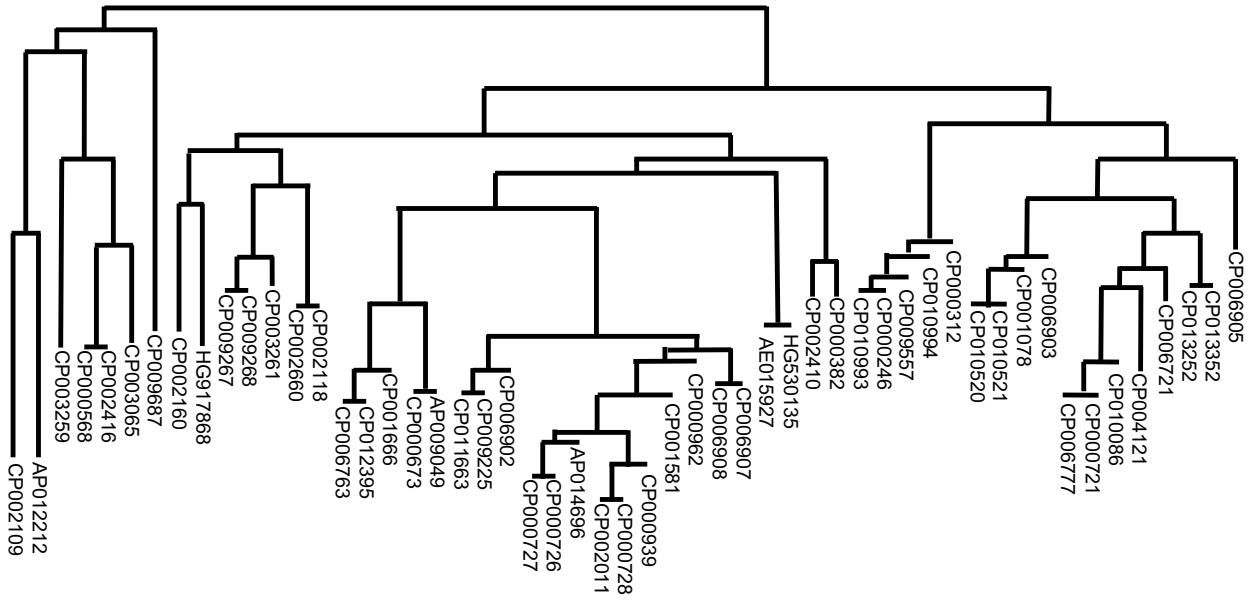


Figure 2: The phylogeny of the *Clostridium* genomes considered (represented by their NCBI accession number). The pairs of genomes sharing a parent are considered close on the evolutionary scale and have a recent common ancestor.

player's solution is worse than the top score by more than two moves (*i.e.* the score > (top score+2)), the player loses progression points, and otherwise, they win progression points. This tolerance range of two moves above the top score was used to not penalize players on puzzles for which the top score represents a potential optimal solution that cannot be improved and is hard to achieve. The player gets promoted to the next difficulty level upon reaching a predetermined number of progression points, which was set to 30 during the period of this study. If the player has a negative sum of progression points, they are demoted to the previous difficulty level, if possible. The system was set up so that the progression points gained (+10) are twice the amount of points lost (-5), so that it is easier to get promoted than to get demoted. In other words, this means that a player could get promoted after solving three puzzles in a row with a good score (within the set tolerance range).

This system was designed to ensure that the player has access to puzzles that are compatible with their level of experience in the game, keeping them in the flow channel [24, 39]. If the player is promoted when already at the last difficulty level, the player gains a *prestige level* and returns to the first difficulty level. After this, the player continues playing with the same progression system, except that they gain bonus leaderboard points according to their level of prestige. This allows the most experienced players to produce solutions again for the puzzles of lower difficulty.

the game was announced on Twitter[4] and through SciStarter[5], a website specialized in promoting citizen science projects. Participation was voluntary and anonymous; the only action required from the player was to read and accept the consent form, and choose a username and password. No demographic data were collected; only information from the matches was saved in the database, where a match corresponds to one puzzle solved by one player. The data saved for each match was the puzzle id, the player's id/username, the final score obtained, the list of moves used by the player, the total time spent on solving the puzzle, the time and date of the match completion, and if the player restarted the match or not.

The two authors considered all the 1,197 matches completed by the 116 distinct players registered during the study period, and focused on further analyzing the 278 completed matches on real genome puzzles. This analysis was based on first classifying the quality of the solutions as determined by a comparison of the score with the top score of a puzzle. Then, the list of moves was analyzed to better understand player strategies that lead to optimal or sub-optimal solutions, which characteristics made puzzles more difficult for the players and how the dynamic difficulty adjustment was having an effect on progression. For all the statistical tests reported below, an alpha level of 0.05 was used. The following section describes the coefficients that we defined to characterize the location of the events in the puzzles and the ratio used to evaluate solution quality.

4 METHODOLOGY

4.1 Study Design

For this paper, data collected between August 2021 and August 2022 were analyzed. As a strategy to recruit participants, the release of

[4]https://twitter.com/
[5]https://scistarter.org/

4.2 The Alpha and Delta Coefficients and the S/TS Ratio

As mentioned before (section 3.1), the puzzles in the game are first presented with the terminus in the middle of the screen (Figure 1). Initially, it is possible to see about ten genes on each side of the terminus. When moving the view while still keeping the terminus visible on the screen, it is possible to see at least 20 genes on either side of it. For this reason, we used a range of 20 genes on each side of the terminus (40 genes in total) to determine if events are located near the terminus (within 20 genes on each side) or far from it.

In order to get a measure of where players are finding events on each puzzle, we defined the following coefficients. We first calculated the *alpha coefficient* for each player solution on a puzzle, which is simply a sum of scores given to each move performed by the player. Moves are scored based on their length (number of genes) and multiplied by -1 when near the terminus, or +1 when far from the terminus. Then, for each puzzle, we calculate the *delta coefficient* as the average of the alpha coefficients of the respective puzzle, but only for solutions that achieved a score equal to or lower than the top score. This filtering was done to avoid biasing the delta coefficient with events that constitute sub-optimal solutions. As a result, a delta coefficient that is less (respectively greater) than 0 indicates that the majority of the genes affected by the moves are located near (respectively far from) the terminus.

We then defined the *S/TS ratio* to evaluate the quality of a puzzle solution. This ratio is the number of moves used by a player to solve a puzzle (score), divided by the number of events calculated by the algorithm (top score). A ratio of 1 indicates that a human solution is on par with the algorithm solution, and a ratio that is less (respectively greater) than 1 indicates a strong (respectively weak) solution obtained by the player.

5 RESULTS

5.1 Analysis of players' strategies

Of the 64 puzzles created based on genomes, only 30 were played and completed. This can be explained in part by the fact that none of the players managed to pass difficulty level 6 (out of the eight available) during the evaluated period. Players beat the top score in 5 puzzles. Two of these had a low top score (2), two were at difficulty level 5 and the last one at difficulty 6. More details on these puzzles are presented in Table 1. We checked all the players' strategies on all genomic puzzles to understand and learn how they were approaching the puzzles and what led to good or bad solutions.

One of the main winning strategies involved using inversions when the algorithm did not. Inversions, which are more difficult to spot, have the potential to reduce considerably the total number of events required to solve a puzzle. Indeed, at least two duplications (copying genes to the other side of the terminus) and two deletions (removing the sources of the duplications) would be required to reproduce an inversion event around the terminus region. That alone would be equivalent to a score of 4 instead of a score of 1. We observed the use of inversions in 3 out of the five cases where users beat the top score. In two of them, the use of an inversion helped the players find a solution with six moves in a problem with a top score of 11. In another case, the player solved a puzzle with a top score of 25 in 19 moves because of an inversion. Of course, inversions are not always going to be useful in all puzzles, but the ability of players to identify these patterns are a great way to improve on solutions.

Another important observation is that in the vast majority of games (about 96%) where players achieved a score at least as good as the top score, players started to solve the puzzles from the terminus region and then worked their way towards the extremities. Since both sides of the genomes are anchored on the terminus, the alignments of the genes are always going to start next to the terminus. For example, a deletion operation will always move the leftover genes towards the terminus to fill the gap. As a result, focusing on the inside of the genomes last could potentially lead to breaking some alignments in the extremities that seemed to match previously.

We also looked for player strategies that could lead to worse solutions. One of the main ones was related to the focus on smaller, more local, moves to make a match rather than looking for a more global move that could potentially fix two mismatches at once. Two different examples of this are presented in Figures 3 and 4. Figure 3-A presents a sub-optimal solution where a player first duplicated the pink triangle to create one match, and then had to delete three contiguous genes further to the left-hand side to complete the puzzle. A more global approach, presented in Figure 3-B, can complete the puzzle by deleting the pink circle and triangle instead, which creates a perfect alignment of the remaining genes. A different example of this sub-optimal local strategy is shown in 4-A, where a player deleted the two white genes first to create a match for the pink hexagon and star, and then had to delete all the remaining genes to complete the puzzle. A more global strategy, shown in Figure 4-B, chooses to match a different pair of pink hexagon and star (the farthest one to the right) and this allows to solve the puzzle using only one large deletion.

5.2 Location of events

When analyzing players' solutions, we noticed that some puzzles with similar genome sizes and top scores were more complex than others. One hypothesis for this was that puzzles in which the majority of the genes affected by evolutionary events are located far from the terminus (where the genomes are anchored) might be more difficult than the others.

When looking at the delta coefficients (see section 4.2) of the different real genome puzzles played, we noticed that almost two thirds of the completed puzzles (18) had what we call a *balanced* delta coefficient, with values in the range of $[-3, 3]$. Note that there were only 2 puzzles with delta coefficients < -3, so we excluded them from this analysis. The remaining completed puzzles (10) had delta coefficients ≥ 6, exhibiting more of an imbalance towards having more genes located far from the terminus that are affected by events (we called this group of puzzles *imbalanced*). We analyzed the performance of players using the S/TS ratios for both the balanced and imbalanced groups of puzzles, and found that the average of the S/TS ratios was higher in the imbalanced set of puzzles (2.61) compared to the balanced set (1.71). Since values of S/TS ratios were not normally distributed (the Shapiro-Wilk test rejected the null hypothesis of a normal distribution with $W = 0.36, p < .001$),

Table 1: Details of puzzles in which players found better solutions than an algorithm. The *id* is the unique identifier for each puzzle. *Mutable* is the name of the genome the player can manipulate, and *Target* is the name of the reference genome. *Difficulty* is the puzzle's difficulty level. *Plays* indicates how many times that puzzle was played during the evaluation period. *Wins* shows the number of times a player produced a better solution than the Top Score. *Score* is the optimal number of moves found by players and *Top Score* shows the number of events calculated by the OrthoAlign algorithm.

id	Mutable	Target	Difficulty	Plays	Wins	Score	Top Score
60e635aa7765e86e5cff886b	AP014696	CP000726	1	17	11	1	2
60e635b97765e86e5cff8871	AP014696	CP000727	2	12	9	1	2
60e637fc7765e86e5cff88ef	CP001078	CP006903	5	1	1	6	11
60e637f67765e86e5cff88ec	CP006903	CP001078	5	1	1	6	11
60e62ef27765e86e5cff880b	CP002160	HG917868	6	1	1	19	25

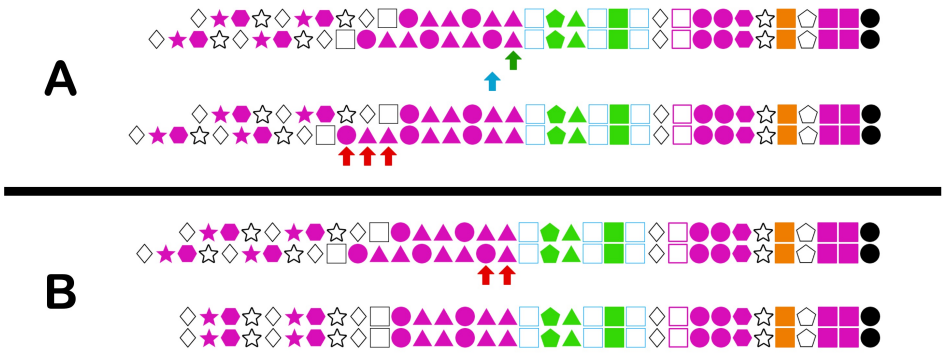


Figure 3: id: *60e636c97765e86e5cff88b9*. Target (1st row): *CP000382*. Mutable (2nd row): *CP002410*. In *A*, the green arrow marks the duplication source, the blue arrow marks its destination, and the red arrows represent deletions. The player opted for a local solution, duplicating first and then removing the surplus. In *B*, the player could see a better solution to the same problem with just one move.

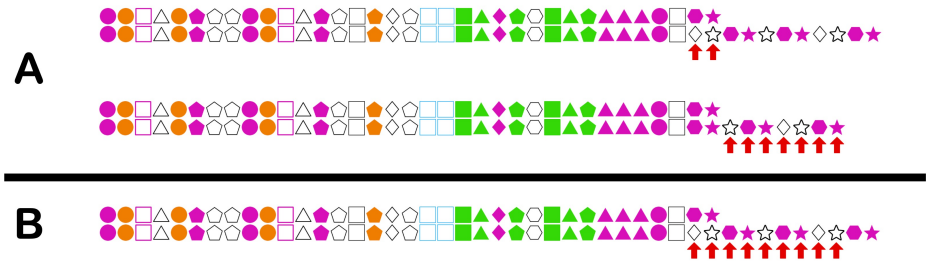


Figure 4: id: *60e635cd7765e86e5cff887a*. Target (1st row): *CP002011*. Mutable (2nd row): *CP000728*. In *A*, the player opted for a local solution, removing a pair of genes (indicated by red arrows) to complete the alignment and then removing the surplus. In *B*, the player could see a better solution to the same problem with just one move.

we performed a Mann-Whitney U test (see Figure 5 for a summary of the results). The test showed a statistically significant difference of S/TS ratios between the balanced ($N = 183$) and imbalanced ($N = 91$) groups ($z = -3.74, p < .001$), rejecting the null hypothesis that the S/TS ratios were sampled from the same distribution. This result suggests that the puzzles in which the majority of the genes affected by events are far from the terminus are harder to solve than those with a more balanced distribution of event locations.

5.3 Influence of patterns on player performance

Our analysis found that in certain puzzles where players struggled to get an S/TS ratio equal to 1, there were tandemly repeated clusters of genes. Figure 6-A shows one example of this, where some players mixed up which group of genes to remove, choosing to delete the group of genes with an orange circle instead of removing the group that contained the orange square. Figure 6-B shows one of the possible solutions with an S/TS ratio equal to 1, found by another player.

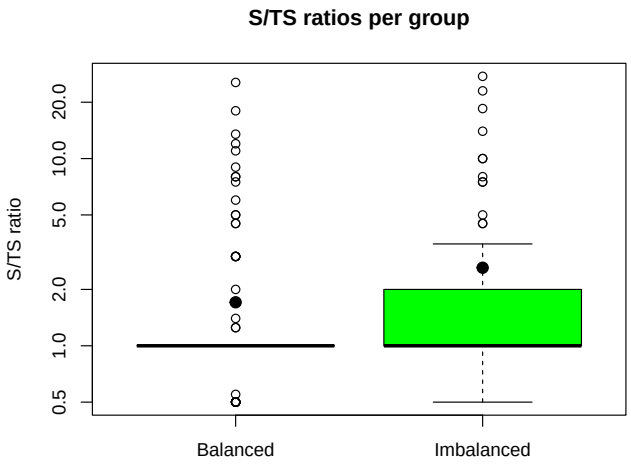


Figure 5: A boxplot of the S/TS ratios for the balanced and imbalanced groups of puzzles. The filled black dots are the means. A Mann-Whitney U test shows a significant difference between the groups, $z = -3.74, p < .001$**.**

The puzzle shown in Figure 6 had an S/TS ratio greater than one for 37% of matches played. In its mirrored version, this was observed in 45% of games, which indicates that these were difficult puzzles for players. It seems that inattention blindness or optical illusions could be responsible for this, since the repetitive pattern of 3 green shapes and 1 orange shape made it difficult for players to differentiate the orange circles and squares. It's also possible that the players were scanning this area of the puzzle from left to right (F-shaped pattern), which made it seem like the extra genes that had to be deleted were on the far left of the pattern. Moreover, since the square and circle shapes both occupy a similar area (in addition to being the same color), it made it easier to assume, at first glance, that they might be the same.

5.4 Dynamic Difficulty Adjustment

In section 3.3, we presented a simple system that adjusts the difficulty dynamically in GeSort. Remember that the proposed method does not just promote the players when successful; it keeps them at the same level or demotes them if necessary to guarantee that they stay in the flow channel. To analyze its effectiveness, we averaged the S/TS ratio of all puzzles played at each difficulty level (Figure 7).

In earlier levels, and more specifically the first two levels, the S/TS ratio is significantly higher. This suggests that players in the first two difficulty levels tend to struggle more, on average, with the puzzles. This makes a lot of sense, since the players who struggle the most are mainly going to play in difficulty level 1 and occasionally be promoted to level 2, before being sent back to level 1. Interestingly, as the difficulty level increases, the average S/TS ratio decreases consistently, which suggests that players who progressed through several difficulty levels were prepared for more challenging games. A Pearson correlation coefficient was computed to assess the linear relationship between the average of the S/TS ratio and the difficulty level. There was a negative correlation between the two variables, $r^2 = 0.8571, r(4) = -0.9258, p = 0.008$. This indicates that the system is effective at making strong players progress and maintain weaker players in the first few levels of difficulty, where they can practice and improve.

6 DISCUSSION AND CONCLUSION

The GeSort citizen science game differs somewhat from its peers in the bioinformatics domain because the solution produced by players is a sequence of events, and the order of these events can influence the final result. For this reason, a considerable focus of the research on GeSort is about analyzing and learning from the strategies used by players. The data analyzed here showed that some players could devise more efficient strategies for sorting genomes compared to the results obtained with a heuristic. The observations made in this paper constitute a first step towards the ultimate goal of improving algorithms for the genome sorting problem. One of the challenges in GeSort's design was presenting puzzles for large genomes on the screen. Displaying only portions of the genomes was never a possibility, since it might remove genes that could be sources of duplications for example. Changing the size of the genes to fit entire genomes on the screen would have been overwhelming and could cause more errors as smaller shapes would be harder to differentiate. The compromise that we used, as can be seen in Figure 1, is to provide a standard view displaying roughly 20 genes at a time and the option to show, at the same time, a mini-map of the entire genomes. Another feature that we added to the mini-map is navigation. When clicking on any gene of the mini-map, the view on the main screen is automatically centered around the selected gene. In the end, our results show that puzzles are harder to solve when the majority of the genes affected by events are far from the terminus. We believe that since the genomes are anchored on the terminus, being able to see the terminus on the screen serves as a reference point which helps identify the matched genes and the best moves. As players work on parts of the genomes that are far away from the terminus, they might lose this reference point which can cause more errors.

Our results have shown that repetitive patterns in puzzles made up of different shapes and colors can lead to some mistakes. Moreover, some shapes of the same color can be mixed up when surrounded by these repetitive patterns. In GeSort, the shapes and colors were randomly distributed among the 24 possible genes that could appear in the dataset. Some puzzles, by chance, might contain some patterns that are more difficult to see for players. Based on our analysis, if similar patterns and repetitive regions are predicted to appear consistently in a puzzle game, we propose as a design guideline that *color and shape assignments be made carefully as to maximize the visual differences between these similar patterns.* Whenever there exists different patterns but they are not generalizable to all puzzles (such as in GeSort), it might be preferable to reassign colors and shapes individually for each puzzle as an automated pre-processing step.

In the same vein, for puzzle games that rely on matching items, we propose that *visual cues should be used to help users identify and confirm matches.* This would have been useful especially for the

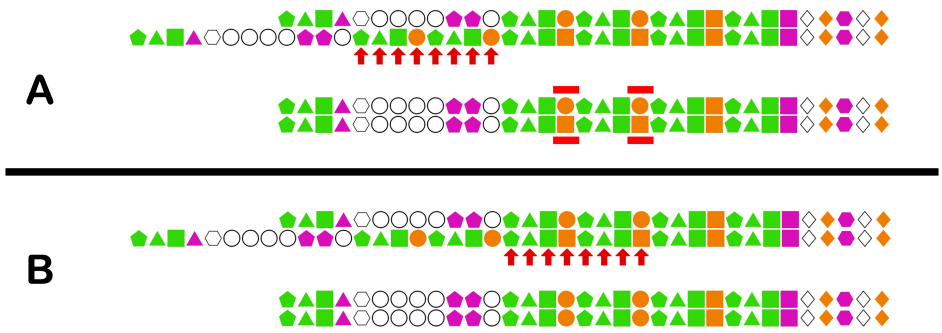


Figure 6: id: *60e635aa7765e86e5cff886b*. Target (1st row): *CP000726*. Mutable (2nd row): *AP014696*. In *A*, the red arrows indicate the genes selected for deletion. There are five groups of 4 genes and two of them must be removed. The player deletes the ones with the orange circle instead of those with the orange square, resulting in an incorrect alignment. *B* shows a valid solution in which the correct group of genes is deleted.

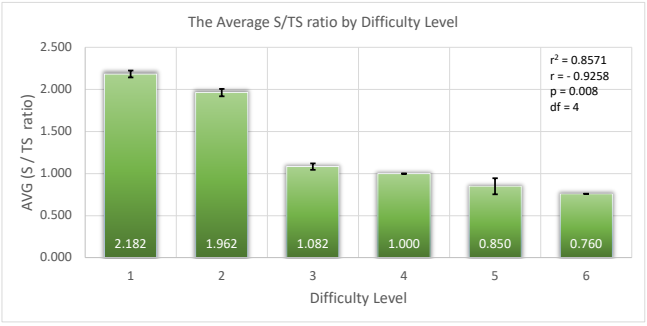


Figure 7: Each bar represents the average S/TS ratio of puzzles at the same difficulty level. The error bar represents the standard error. There was a negative correlation between the average S/TS ratio and the difficulty level, $r^2 = 0.8571$, $r(4) = -0.9258, p = 0.008$.

case presented in Figure 6. An indication by the game that there was a mismatch between the orange circle and square would have allowed the players to quickly undo the most recent move and find a better solution. While it is simple to provide a visual cue for matches in regions where the genes are aligned (on top of each other), one could also consider to indicate where matches could be when the genes are not aligned. Some visual cues like this are common in puzzle games where the player needs to find patterns and match pieces. In the game Candy Crush Saga[6], where the player must combine three or more elements of the same color and shape, the game can give a hint showing a possible move. However, this type of help should be used sparingly, especially in the context of a human computation game. A hint in this context would have to be inferred from an algorithm, which could bias the player solutions towards either a non-optimal solution (local move) or a solution that is similar to the ones produced by algorithms (which we already have access to).

In terms of dynamic difficulty adjustment, the system implemented in GeSort, although simple, proved to be effective. We believe the proposed system can work for any puzzle game, as long as the value of a potentially suitable solution (top score) is available. Moreover, in addition to being an essential part of adjusting the difficulty, the presence of an achievement goal can also increase engagement [31]. Besides being straightforward to implement, the system is also easily customizable, requiring only to change the progression score threshold or the number of progression points gained/lost to make adjustments. Another benefit of the system is that since the puzzles are always randomized within each difficulty level, players cannot easily notice when they are demoted, thus avoiding possible decreases in engagement. The potential downside is that a fraction of players can get stuck at the initial difficulty levels, producing an imbalance in the number of solutions obtained for different levels. However, only promoting players to higher difficulty levels when they are ready guarantees more optimal solutions to the complex puzzles, as demonstrated by our results.

6.1 Limitations and Future Work

As the GeSort game was initially released with the dynamic difficulty adjustment system, we were not able to compare it directly with a control condition that did not have it. In future work, it would be interesting to further evaluate the DDA system proposed here by testing it with different thresholds for promotion/demotion and a control.

As mentioned above, one of the complexities of designing the GeSort game was to find the right balance regarding the quantity of information that we can display on the screen. Our approach was to make a compromise between the size of each gene (so they can be distinguishable) and the total size of the genomes (showing as much of them as possible), along with a mini-map that provides a more global view and a way to quickly navigate the genomes. An untested alternative would be to allow the player to select some genes that appear to be frequently clustered together and collapse them, along with the ability to expand these collapsed blocks if necessary to gain access to the genes they contain (e.g. for duplications). This would allow the players to effectively reduce the total amount of information presented to them.

[6]https://www.king.com/game/candycrush

Finally, GeSort is currently only available as a web application. In future work, we also aim to develop and release a mobile version of GeSort, and study how the platform affects the solutions produced by players. Using our first observations presented in this paper along with all the aggregated solutions obtained on all platforms, we will then aim to develop new algorithmic strategies for the genome sorting problem.

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